On semi-relaxed optimal transport

Maxime Nicaise, Yaiza Bermudez, and Samir M. Perlaza

RESEARCH REPORT
N° 9607
January 2026
Project-Team NEO

ISSN 0249-6399 ISRN INRIA/RR--9607--FR+ENG

On semi-relaxed optimal transport

Maxime Nicaise, Yaiza Bermudez, and Samir M. Perlaza

Project-Team NEO

Research Report n° 9607 — January 2026 — 30 pages

Abstract:

Key-words: Optimization, Optimal Transport, Entropic Optimal Transport, Regularized Optimal Transport.

Maxime Nicaise, Yaiza Bermudez and Samir M. Perlaza are with INRIA, Centre Inria d'Université Côte d'Azur, Sophia Antipolis 06902, France. Samir M. Perlaza is also with the GAATI Laboratory at the Université de la Polynésie Française, Faaa 98702, French Polynesia; and also with the Electrical and Computer Engineering Department at Princeton University, Princeton N.J. 08544, USA.

RESEARCH CENTRE
SOPHIA ANTIPOLIS – MÉDITERRANÉE

2004 route des Lucioles - BP 93
06902 Sophia Antipolis Cedex

Sur le transport optimal semi-relaxé

Résumé :

Mots-clés : Optimisation, Transport Optimal, Transport Optimal Entropique, Transport Optimal Régularisé.

Contents

1 Introduction **4**
1.1 State of the art 4
1.2 Notations and preliminaries 4

2 Semi-relaxed optimal transport **5**

3 Sinkhorn-like regularization scheme **5**
3.1 First dual point of view changer le titre ou degager la section . . 6
3.2 Another dual formulation 7
3.2.1 Weak duality 7
3.2.2 Attainment 8
3.3 Primal attainment 15
3.4 (α, β)-approximation error control renommer la section c naze . . 18
3.5 Iterative algorithms 19
3.5.1 Presentation of the algorithms 19
3.5.2 Theoretical guarantees 19
3.6 Experimental results 29

1 Introduction

l'intro est a refaire quasiment en entier pour bien introduire le transport optimal semi relaxé. A mon avis il faut aussi refondre toute l'organisation de ce document.

1.1 State of the art

Intro sur le transport optimal semi relaxe.

1.2 Notations and preliminaries

Il y a des notations aajouter et d'autres à enlever. Given a function $f : \mathcal{X} \to \mathcal{Y}$, its effective domain is denoted by $\mathrm{dom}(f) \triangleq \{x \in \mathcal{X} : f(x) < +\infty\}$; and its range is denoted by $\mathrm{Ran}(f) \triangleq \{f(x) : x \in \mathcal{X}\}$. Given a set $\mathcal{A}$, its interior is denoted by $\mathrm{int}(\mathcal{A})$. Given two measures μ and ν, the notation $\mu \ll \nu$ signifies that μ is absolutely continuous with respect to ν. Across this report, $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces. The set of probability measures over $\mathcal{X}$ equipped with the σ-field of Borel sets is denoted by $\triangle(\mathcal{X})$. The set of p-integrable functions with respect to P is denoted by $\mathrm{L}^p(\mathcal{X}, P)$. Given two probability measures $P \in \triangle(\mathcal{X})$ and $Q \in \triangle(\mathcal{Y})$, the notation $\Pi(P, Q)$ denotes the set of probability measures on $\mathcal{X} \times \mathcal{Y}$ with marginals P and Q in $\triangle(\mathcal{X})$ and $\triangle(\mathcal{Y})$, respectively. The probability measures in $\Pi(P, Q)$ are called couplings. By extension, $\Pi(P, \star)$ denotes the set of probability measures on $\mathcal{X} \times \mathcal{Y}$ where only the first marginal is constrained to be P. The product measure of $P \in \triangle(\mathcal{X})$ and $Q \in \triangle(\mathcal{Y})$ in $\triangle(\mathcal{X} \times \mathcal{Y})$ is denoted by PQ. The first marginal of a measure $\pi \in \triangle(\mathcal{X} \times \mathcal{Y})$ is denoted by $\pi^{\mathcal{X}}_{\#}$. Similarly, its second marginal is denoted by $\pi^{\mathcal{Y}}_{\#}$. Given two functions $f : \mathcal{X} \to \mathbb{R}$ and $g : \mathcal{X} \to \mathbb{R}$, the notation $f \overset{a.s.}{\underset{P}{=}} g$ means that f and g are almost surely equal with respect to P. Similarly, the notation $f \overset{a.s.}{\underset{P}{\geq}} g$ means that f is greater than g almost surely with respect to P. Given $\mathcal{A}$ and $\mathcal{B}$, two subsets of $\mathcal{X}$, the notation $\mathcal{A} \overset{a.s.}{\underset{P}{\subset}} \mathcal{B}$ means that $P(\mathcal{A} \setminus \mathcal{B}) = 0$. Let $\mathcal{E}$ be a Banach space and $\mathcal{E}'$ be its topological dual space. Given $x \in \mathcal{E}$ and $y \in \mathcal{E}'$, $\langle y, x \rangle$ denotes the dual pairing between y and x. If $f : \mathcal{X} \to \mathbb{R}$ and $g : \mathcal{Y} \to \mathbb{R}$ are two real functions then $f \oplus g : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is the function defined through $f \oplus g(x, y) = f(x) + g(y)$. The following definition formalizes the notion of admissibility for an optimization problem, ensuring that candidates yield a finite value within the functional domain.

Definition 1.1. *Given a set $\mathcal{M}$ and a functional $R : \mathcal{M} \to \mathbb{R} \cup \{+\infty\}$, an instance $\theta \in \mathcal{M}$ is said to be admissible for the following problem*

$$\inf_{\nu \in \mathcal{M}} R(\nu), \tag{1}$$

if $R(\theta) < +\infty$.

Analysis of optimality conditions and the derivation of dual problems rely heavily on the transformation of functions into a dual space representation. This operation is performed through the convex conjugate, providing a bridge between primal and dual variables in Banach spaces. The convex conjugate is defined below.

Definition 1.2. *Let $\mathcal{E}$ be a Banach space and $\mathcal{E}'$ be its topological dual space. Let $\phi : \mathcal{E} \to \mathbb{R} \cup \{-\infty, +\infty\}$; the convex conjugate (or Fenchel conjugate) of ϕ is the function $\phi^* : \mathcal{E}' \to \mathbb{R} \cup \{-\infty, +\infty\}$ defined for $y \in \mathcal{E}'$ through (see [1, Section 2.3])*

$$\phi^*(y) = \sup_{x \in \mathcal{E}} (\langle y, x \rangle - \phi(x)). \tag{2}$$

Complementary to the convex conjugate, the indicator function serves as a fundamental tool through which constraints are encoded directly into the objective functional. This formulation is particularly suited for optimal transport problems, as it ensures the strict enforcement of marginal constraints. The following definition formalises this concept, providing a mechanism to assign an infinite cost to any instance outside the feasible set.

Definition 1.3. *Let $\mathcal{C}$ be a subset of a set $\mathcal{E}$. The indicator function of $\mathcal{C}$, denoted through $\iota_{\mathcal{C}}$, is defined as:*

$$\iota_{\mathcal{C}}(x) = \begin{cases} 0 & \text{if } x \in \mathcal{C}; \\ +\infty & \text{if } x \notin \mathcal{C}. \end{cases} \tag{3}$$

2 Semi-relaxed optimal transport

Throughout this section, $(\mathcal{X}, P)$ and $(\mathcal{Y}, Q)$ are fixed probability spaces. The function $\mathsf{c} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a lower semi-continuous measurable function. Consider the problem of semi-relaxed optimal transport, introduced in , defined as follows

$$\inf_{\pi \in \Pi(P, \star)} \int_{\mathcal{X} \times \mathcal{Y}} \mathsf{c}(x, y) \mathrm{d}\pi(x, y). \tag{4}$$

Relevant literature concerning this formulation includes ..... Standard approaches for solving semi-constrained optimal transport problems often rely on linear programming. However, these methods involve significant computational complexity, which frequently becomes prohibitive for large-scale datasets or high-dimensional problems. trouver une ref si possible pour le semi constrained qui traite le pb avec une solution de programmation lineaire. Expliquer que c'est pas bien d'un point de vue computationnel.

3 Sinkhorn-like regularization scheme

Throughout this section, $(\mathcal{X}, P)$ and $(\mathcal{Y}, Q)$ are fixed probability spaces. The function $\mathsf{c} : \mathcal{X} \times \mathcal{Y} \to \mathbb{R}$ is a lower semi-continuous, measurable, bounded function, and α and β are strictly positive numbers. The objective is to extend

the results established in [2] quand meme etendre un peu la biblio associee a sinkhorn to the framework of semi-relaxed optimal transport. This goal is achieved through a specific choice of regularization that ensures the dual problem is amenable to Sinkhorn-like iterations. The proposed scheme fits within the general framework introduced in [3]. More precisely, the problem is a particular instance of [3, Equation 3.1] where $n = 1$, the functional $\mathcal{F}_1$ is the indicator function of the set $\{P\}$, and $\mathcal{F}_2$ equals β times the Kullback-Leibler (KL) divergence with respect to Q, with β strictly positive. The reference measures, denoted as dx and dy in [3], correspond here to P and Q, respectively. This specialization leads to the following problem

$$\inf_{\pi \in \Pi(P,\star)} \mathsf{P}(\pi) \triangleq \int_{\mathcal{X}\times\mathcal{Y}} \mathsf{c}(x,y)\mathrm{d}\pi(x,y) + \alpha D_{\mathrm{KL}}(\pi||PQ) + \beta D_{\mathrm{KL}}(\pi^{\mathcal{Y}}_{\#}||Q). \quad (5)$$

3.1 First dual point of view changer le titre ou degager la section

From [3, Theorem 3.2], the dual of problem (5) can be written as follows

$$\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta \int_{\mathcal{Y}} \left(e^{-\frac{v(y)}{\beta}} - 1\right) \mathrm{d}Q(y)$$
$$-\alpha \int_{\mathcal{X}\times\mathcal{Y}} \left(e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}} - e^{-\frac{\mathsf{c}(x,y)}{\alpha}}\right) \mathrm{d}PQ(x,y). \quad (6)$$

The theoretical results from [3] remain applicable to this dual formulation. In particular, strong duality holds, ensuring equality between the supremum in (6) and the infimum in (5). The convergence properties established in [3, Proposition 3.9, Theorem 4.1] are also valid. Nevertheless, this work aims to provide enhanced convergence results and refined theoretical guarantees. From [3, Section 3.3, Table 1], the iterates obtained by alternating maximization on D in (6) can be written as follows

$$u^{(l+1)} = -\alpha \log\left(\int_{\mathcal{Y}} e^{\frac{v^{(l)}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y)\right); \quad (7a)$$

$$v^{(l+1)} = -\frac{\alpha\beta}{\alpha+\beta} \log\left(\int_{\mathcal{X}} e^{\frac{u^{(l+1)}(x)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}P(x)\right). \quad (7b)$$

Or alternatively by considering the so-called scaled iterations

$$(a^{(l)}, b^{(l)}) \triangleq (e^{\frac{u^{(l)}}{\alpha}}, e^{\frac{v^{(l)}}{\alpha}}), \quad (8)$$

for which the algorithm is written as follows

$$a^{(k+1)}(x) = \left(\int_{\mathcal{Y}} e^{-\frac{\mathsf{D}(x,y)}{\alpha}} b^{(k)}(y)^{-\frac{\beta}{\alpha}} \mathrm{d}Q(y)\right)^{-1}; \quad (9a)$$

$$b^{(k+1)}(y) = \left(\int_{\mathcal{X}} e^{-\frac{\mathsf{D}(x,y)}{\alpha}} a^{(k+1)}(x) \mathrm{d}P(x)\right)^{\frac{\alpha}{\alpha+\beta}}. \quad (9b)$$

Dire qu'on peut faire quand meme un peu mieux surtout theoriquement que ces resultats qui ont ete etablis dans un cadre bien + general

3.2 Another dual formulation

This section proposes an alternative dual formulation aiming to strengthen the convergence results of the potentials without requiring finite space assumptions. The derivation in [3] is highly general and does not explicitly account for the fact that the fasible couplings for the primal problems are necessarily probability measures. The alternative derivation presented here utilizes the variational formula of the KL divergence. The dual problem is subsequently obtained through the inversion of the infimum and supremum. This approach results in a formulation situated in the log-domain, which shares structural similarities with the notion of free energy in statistical physics citer des refs pour cette idee

$$\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \triangleq \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta \log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right) \\ -\alpha \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right). \quad (10)$$

3.2.1 Weak duality

Lemma 3.1. *Weak duality holds between the primal* (5) *and the dual problem* (10)*, that is*

$$\sup_{u\in \mathrm{L}^\infty(\mathcal{X}), v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \leq \inf_{\pi\in\Pi(P,\star)} \mathsf{P}(\pi) < +\infty. \quad (11)$$

Proof: Let $\pi \in \Pi(P,\star)$ be any feasible measure for problem (5); in particular π is absolutely continuous with respect to the product measure PQ and $\pi^{\mathcal{X}}_{\#} = P$. Let $(u,v) \in \mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$ be arbitrary potentials. The Fenchel-Young inequality for the function $a \mapsto \alpha(a\log(a) - a + 1)$ states

$$\alpha(a\log(a) - a + 1) \geq at - \alpha(e^{t/\alpha} - 1) \geq at - e^{t/\alpha}, \quad (12)$$

for all $t \in \mathbb{R}$ and $a > 0$. Let (x,y) be in $\mathcal{X}\times\mathcal{Y}$; applying the previous inequality with $a = \frac{\mathrm{d}\pi}{\mathrm{d}PQ}(x,y)$ and $t = u(x) + v(y) - \mathsf{c}(x,y)$ and integrating with respect to PQ yields

$$\alpha D_{\mathrm{KL}}(\pi||PQ) \geq \int_{\mathcal{X}\times\mathcal{Y}} (u(x) + v(y) - \mathsf{c}(x,y))\mathrm{d}\pi(x,y) \\ -\alpha \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right). \quad (13)$$

Similarly, the following inequality can be deduced from (12)

$$\beta D_{\mathrm{KL}}(\pi^{\mathcal{Y}}_{\#}||Q) \geq \int_{\mathcal{Y}} -v(y)\mathrm{d}\pi^{\mathcal{Y}}_{\#}(y) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right). \quad (14)$$

Addition of the primal cost term $\int \mathsf{c}(x,y)\mathrm{d}\pi(x,y)$ to the sum of these inequalities leads to

$$\mathsf{P}(\pi) \geq \int_{\mathcal{X}\times\mathcal{Y}} (u(x)+v(y)-\mathsf{c}(x,y))\mathrm{d}\pi(x,y) + \int_{\mathcal{X}\times\mathcal{Y}} \mathsf{c}(x,y)\mathrm{d}\pi(x,y) - \int_{\mathcal{Y}} v(y)\mathrm{d}\pi^{\mathcal{Y}}_{\#}(y)$$
$$-\alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right). \tag{15}$$

The definition of the marginals and the constraint $\pi \in \Pi(P,\star)$ imply the identity

$$\int_{\mathcal{X}\times\mathcal{Y}} (u(x)+v(y))\mathrm{d}\pi(x,y) = \int_{\mathcal{X}} u(x)\mathrm{d}P(x) + \int_{\mathcal{Y}} v(y)\mathrm{d}\pi^{\mathcal{Y}}_{\#}(y). \tag{16}$$

Substitution of this identity into the previous expression results in the cancellation of terms involving c and v. The remaining terms satisfy

$$\mathsf{P}(\pi) \geq \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right)$$
$$-\beta\log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right), \tag{17}$$

where the right-hand term is exactly $\mathsf{D}(u,v)$. This inequality holds for all u,v and all feasible π, thus confirming $\sup \mathsf{D}(u,v) \leq \inf \mathsf{P}(\pi)$. Regarding the finiteness of the two values, note that from the fact that c is bounded and PQ is in $\Pi(P,Q)$ and satisfies $D_{\mathrm{KL}}(PQ||PQ) = 0$, it holds that $\mathsf{P}(PQ) < +\infty$. ■

faire la remqrque qu'en particulier cela implique que la valeur du dual est finie

3.2.2 Attainment

Dire que cette formulation duale, bien que très différente en apparence, donne une version à peine différente des itérations par maximisation alternée. La propriete suivante a l'air un peu inutile et en plus elle traduit le fait que la fonctionnelle D n'est pas strictement convexe, donc il n'y aura pas unicité des potentiels maximisants, mais en fait elle va etre utile pour prouver l'existence par contre de potentiels.

Lemma 3.2. *For all functions $u \in \mathrm{L}^\infty(\mathcal{X}), v \in \mathrm{L}^\infty(\mathcal{Y})$ and real numbers M, N, the dual functional D in (10) satisfies*

$$\mathsf{D}(u,v) = \mathsf{D}(u+M,v) = \mathsf{D}(u,v+N). \tag{18}$$

Proof: Let u and v be two functions in $\mathrm{L}^\infty(\mathcal{X})$ and $\mathrm{L}^\infty(\mathcal{Y})$ respectively,

and M, N be two real numbers. From (10), it follows that

$$\begin{aligned}
\mathsf{D}(u+M,v) &= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) + M - \beta \log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right) \\
&\quad -\alpha \log\left(e^{\frac{M}{\alpha}}\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) && (19)\\
&= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta \log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right) \\
&\quad -\alpha \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right). && (20)\\
&= \mathsf{D}(u,v), && (21)
\end{aligned}$$

which proves the first equality in (18). The same computation can be done to prove the second one, as follows

$$\begin{aligned}
\mathsf{D}(u,v+N) &= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta \log\left(e^{-\frac{N}{\beta}}\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right) \\
&\quad -\alpha \log\left(e^{\frac{N}{\alpha}}\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) && (22)\\
&= \int_{\mathcal{X}} u(x)\mathrm{d}P(x) - \beta \log\left(\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)\right) + N \\
&\quad -\alpha \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) - N. && (23)\\
&= \mathsf{D}(u,v), && (24)
\end{aligned}$$

which concludes the proof of (18). ■

Lemma 3.3. *The dual functional* D *in* (10) *is differentiable on* $L^\infty(\mathcal{X})\times L^\infty(\mathcal{Y})$. *Moreover, for all* $(u,v) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$ *and directions* $(a,b) \in L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, *its partial derivatives are given by*

$$\langle \frac{\partial \mathsf{D}(u,v)}{\partial u}, a\rangle = \int_{\mathcal{X}} a(x)\left(1 - \frac{1}{Z(u,v)}\int_{\mathcal{Y}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}Q(y)\right)\mathrm{d}P(x); \qquad (25)$$

$$\langle \frac{\partial \mathsf{D}(u,v)}{\partial v}, b\rangle = \int_{\mathcal{Y}} b(y)\left(\frac{e^{-\frac{v(y)}{\beta}}}{\int_{\mathcal{Y}} e^{-\frac{v(y)}{\beta}}\mathrm{d}Q(y)} - \frac{1}{Z(u,v)}\int_{\mathcal{X}} e^{\frac{u(x)+v(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}P(x)\right)\mathrm{d}Q(y), \qquad (26)$$

where $Z(u,v) \triangleq \int_{\mathcal{X}\times\mathcal{Y}} \exp\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\mathrm{d}PQ(x,y)$.

Proof: The proof is split into two parts, establishing (25) and (26) respectively. Let (u,v) be in $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, let a be in $L^\infty(\mathcal{X})$, and let b be in $L^\infty(\mathcal{Y})$. Since the functions u, v, a, b are bounded, and exp is continuous, all subsequent differentiations under the integral sign are justified by the dominated convergence theorem [4, Theorem 1.34]. Consider first the derivative with

respect to u. The first term of the functional D in (10) is linear with respect to u and conributes

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \int_{\mathcal{X}} (u(x) + ta(x))\,\mathrm{d}P(x) = \int_{\mathcal{X}} a(x)\,\mathrm{d}P(x), \tag{27}$$

while the term $-\beta \log\big(\int_{\mathcal{Y}} e^{-v/\beta}\,\mathrm{d}Q\big)$ does not depend on u and therefore yields zero when differentiated with respect to u. For the coupling term, differentiation gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} Z(u{+}ta{,}v) = \frac{1}{\alpha}\int_{\mathcal{X}\times\mathcal{Y}} a(x)\,\exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}PQ(x,y). \tag{28}$$

Applying the chain rule for the logarithm yields

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \big(-\alpha \log Z(u+ta,v)\big)\\ &= -\frac{1}{Z(u,v)}\int_{\mathcal{X}\times\mathcal{Y}} a(x)\,\exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}PQ(x,y).\end{aligned} \tag{29}$$

By Fubini's theorem [4, Theorem 8.8], this can be rewritten as

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \big(-\alpha \log Z(u+ta,v)\big)\\ &= -\int_{\mathcal{X}} a(x)\left(\frac{1}{Z(u,v)}\int_{\mathcal{Y}} \exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}Q(y)\right)\mathrm{d}P(x).\end{aligned} \tag{30}$$

Combining (27) and (30) concludes the proof of (25). The differentiation with respect to v is handled analogously. The term $\int_{\mathcal{X}} u\,\mathrm{d}P$ in (10) does not depend on v. For the logarithmic term, differentiation yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0}\left(-\beta\log\int_{\mathcal{Y}} e^{-(v(y)+tb(y))/\beta}\,\mathrm{d}Q(y)\right) = \frac{1}{\int_{\mathcal{Y}} e^{-v/\beta}\mathrm{d}Q}\int_{\mathcal{Y}} b(y)e^{-v(y)/\beta}\mathrm{d}Q(y). \tag{31}$$

The coupling term is handled analogously, by symmetry with respect to the variables u and v, and the result is as follows

$$\begin{aligned}&\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} \big(-\alpha \log Z(u,v+tb)\big)\\ &= -\int_{\mathcal{Y}} b(y)\left(\frac{1}{Z(u,v)}\int_{\mathcal{X}} \exp\!\left(\frac{u(x)+v(y)-\mathsf{D}(x,y)}{\alpha}\right)\,\mathrm{d}P(x)\right)\mathrm{d}Q(y).\end{aligned} \tag{32}$$

Summing (31) and (32) concludes the proof of (26). ■

Ici il faut expliquer qu'on choisit la manière d'écrire "la differentielle vaut 0" mais en imposant une condition de normalisation, d'abord pour obtenir l'unicité, puis aussi parce que la solution du primal s'écrit plus simplement à partir de la solution particulière du dual qui vérifie ces 2 conditions de normalisation. Essayer de prouver un peu les 2 normalisations en question pour pouvoir y faire reference après.

Corollary 3.1. *Let $(\hat{u}, \hat{v})$ be in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$. If for all $x \in \mathcal{X}$, u satisfies*

$$\hat{u}(x) = \Phi(\hat{v})(x); \tag{33}$$

$$\text{with } \Phi(\hat{v})(x) \triangleq -\alpha \log\left(\int_{\mathcal{Y}} e^{\frac{\hat{v}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y)\right), \tag{34}$$

then, $\hat{u}$ maximizes $u \mapsto \mathsf{D}(u, \hat{v})$, where D is in (10). Similarly, if for all $y \in \mathcal{Y}$, v satisfies

$$\hat{v}(y) = \Psi(\hat{u})(y) : \tag{35}$$

$$\begin{aligned}\text{with } \Psi(\hat{u})(y) \triangleq \beta \log&\left(\int_{\mathcal{Y}} \left(\int_{\mathcal{X}} e^{\frac{\hat{u}(x)-\mathsf{c}(x,y')}{\alpha}} \mathrm{d}P(x)\right)^{\frac{\alpha}{\alpha+\beta}} \mathrm{d}Q(y')\right)\\ &-\frac{\alpha\beta}{\alpha+\beta} \log\left(\int_{\mathcal{X}} e^{\frac{\hat{u}(x)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}P(x)\right),\end{aligned} \tag{36}$$

then, $\hat{v}$ maximizes $v \mapsto \mathsf{D}(\hat{u}, v)$. Moreover, if (33) and (35) are simultaneously satisfied, then $(\hat{u}, \hat{v})$ maximizes $(u, v) \mapsto \mathsf{D}(u, v)$.

Proof: The functional D is differentiable and concave on $L^\infty(\mathcal{X}) \times L^\infty(\mathcal{Y})$, and in particular concave with respect to each variable when the other one is fixed. As a consequence, a sufficient condition for optimality is the vanishing of the first-order derivative [1, Chapter 2.4].

Assume that $\hat{v} \in L^\infty(\mathcal{Y})$ is fixed and that $\hat{u}$ satisfies (33). By Lemma 3.3, the first-order derivative of the map $u \mapsto \mathsf{D}(u, \hat{v})$ is given by (25). Injecting (33) into this expression shows that the derivative vanishes identically. By concavity, $\hat{u}$ therefore maximizes $u \mapsto \mathsf{D}(u, \hat{v})$.

Similarly, assume that $\hat{u} \in L^\infty(\mathcal{X})$ is fixed and that $\hat{v}$ satisfies (35). Using again Lemma 3.3, the derivative of $v \mapsto \mathsf{D}(\hat{u}, v)$ is given by (26). Injecting (35) into this expression yields the cancellation of the two terms and thus the vanishing of the derivative. By concavity, $\hat{v}$ maximizes $v \mapsto \mathsf{D}(\hat{u}, v)$.

Finally, if (33) and (35) are satisfied simultaneously, then both partial derivatives vanish. Since $(u, v) \mapsto \mathsf{D}(u, v)$ is jointly concave, $(\hat{u}, \hat{v})$ is a global maximizer of D. ■

la expliquer qu'on commence vraiment le travail qui va permettre d'arriver aux arguments de compacite du meme genre que ceux dans [5], qui sont la cle de l'existence en fait

Lemma 3.4. *For all pair of functions (u, v) in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$, the operators Φ in (34) and Ψ in (36) satisfy the following inequalities*

$$\operatorname{osc}(\Phi(v)) \leq 2\|c\|_\infty; \tag{37}$$

$$\operatorname{osc}(\Psi(u)) \leq \frac{2\beta}{\alpha+\beta}\|c\|_\infty. \tag{38}$$

Proof: The proof is divided in two parts, establishing (37) and (45) respectively. Let v be a function in $\mathrm{L}^\infty(\mathcal{Y})$ and (x, y) be in $\mathcal{X} \times \mathcal{Y}$; the following

inequalities are satisfied

$$-\|\mathsf{c}\|_\infty \leq \quad -\mathsf{c}(x,y) \quad \leq \|\mathsf{c}\|_\infty \tag{39}$$

$$e^{\frac{-\|\mathsf{c}\|_\infty}{\alpha}} e^{\frac{v(y)}{\alpha}} \leq \quad e^{\frac{v(y)-\mathsf{c}(x,y)}{\alpha}} \quad \leq e^{\frac{\|\mathsf{c}\|_\infty}{\alpha}} e^{\frac{v(y)}{\alpha}} \tag{40}$$

$$e^{\frac{-\|\mathsf{c}\|_\infty}{\alpha}} \int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y) \leq \int_{\mathcal{Y}} e^{\frac{v(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y) \leq e^{\frac{\|\mathsf{c}\|_\infty}{\alpha}} \int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y). \tag{41}$$

Taking the logarithm of the previous inequality and multiplying by $-\alpha$ yields

$$-\alpha \log\left(\int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y)\right) - \|\mathsf{c}\|_\infty \leq \Phi(v)(x) \leq -\alpha \log\left(\int_{\mathcal{Y}} e^{\frac{v(y)}{\alpha}} \mathrm{d}Q(y)\right) + \|\mathsf{c}\|_\infty, \tag{42}$$

from which it can be deduced that $\mathrm{osc}(\Phi(v)) \leq 2\|\mathsf{c}\|_\infty$, hence concluding the proof of (37).

To establish (45), let Ψ_1 and Ψ_2 denote the first and second terms of Ψ in (36), respectively. Consider a function $u \in \mathrm{L}^\infty(\mathcal{X})$. The oscillation of the constant part $\Psi_1(u)$ is zero, which implies the identity $\mathrm{osc}(\Psi(u)) = \mathrm{osc}(\Psi_2(u))$. Apart from the probability measure P and the multiplicative factor $\frac{\beta}{\alpha+\beta}$, the functional form of Ψ_2 is symmetric to the definition of Φ in (34) . The proof of (37) establishes that $\mathrm{osc}(\Phi(v)) \leq 2\|\mathsf{c}\|_\infty$. Adoption of an identical reasoning for Ψ_2 with its specific scaling leads to the following bound

$$\mathrm{osc}(\Psi(u)) \leq \frac{2\beta}{\alpha+\beta}\|\mathsf{c}\|_\infty, \tag{43}$$

which concludes the proof of (38). ■

Maintenant qu on a un controle de l oscillation on se rapporche du controe en borne uniforme

Lemma 3.5. *For all pair of functions* (u,v) *in* $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$*, the operators* Φ *in* (34) *and* Ψ *in* (36) *satisfy the following inequalities*

$$\|\Phi(v)\|_\infty \leq \|v\|_\infty + \|c\|_\infty; \tag{44}$$

$$\|\Psi(u)\|_\infty \leq \frac{2\beta}{\alpha+\beta}\|\mathsf{c}\|_\infty. \tag{45}$$

Proof: The proof is divided in two parts, establishing (44) and (45) respectively. Let v be a function in $\mathrm{L}^\infty(\mathcal{Y})$ and (x,y) be in $\mathcal{X} \times \mathcal{Y}$; the following inequalities are satisfied

$$-\frac{\|v\|_\infty + \|\mathsf{c}\|_\infty}{\alpha} \leq \frac{v(y) - \mathsf{c}(x,y)}{\alpha} \leq \frac{\|v\|_\infty + \|\mathsf{c}\|_\infty}{\alpha} \tag{46}$$

$$e^{-\frac{\|v\|_\infty + \|\mathsf{c}\|_\infty}{\alpha}} \leq e^{\frac{v(y)-\mathsf{c}(x,y)}{\alpha}} \leq e^{\frac{\|v\|_\infty + \|\mathsf{c}\|_\infty}{\alpha}}. \tag{47}$$

Succesively integrating the previous inequalities with respect to Q, taking the logarithm and multiplying by $-\alpha$ concludes the proof of (44).

To establish (45), let u be in $\mathrm{L}^\infty(\mathcal{X})$ and note that the definition of Ψ in (36) implies

$$\int_{\mathcal{Y}} e^{-\Psi(u)(y)/\beta} \mathrm{d}Q(y) = 1. \tag{48}$$

As the measure Q has total mass 1, the previous equality implies

$$\inf_{y\in\mathcal{Y}} e^{-\Psi(u)(y)/\beta} \leq 1, \text{ and } \sup_{y\in\mathcal{Y}} e^{-\Psi(u)(y)/\beta} \geq 1. \tag{49}$$

Monotonicity of the logarithm function yields

$$\sup \Psi(u) \geq 0 \quad \text{and} \quad \inf \Psi(u) \leq 0. \tag{50}$$

The interval $[\inf \Psi(u), \sup \Psi(u)]$ contains the origin. The distance from the origin to any point within a closed interval is bounded through the total interval length. The following inequalities result from this inclusion

$$\sup \Psi(u) \leq \operatorname{osc}(\Psi(u)) \quad \text{and} \quad -\inf \Psi(u) \leq \operatorname{osc}(\Psi(u)). \tag{51}$$

The definition $\|\Psi(u)\|_\infty = \max(\sup \Psi(u), -\inf \Psi(u))$ establishes the bound $\|\Psi(u)\|_\infty \leq \operatorname{osc}(\Psi(u))$. Since the oscillation bound in (38) is established as $\frac{2\beta}{\alpha+\beta}\|\mathsf{c}\|_\infty$, this concludes the proof of (45). ■

voila le lemme qui permet de borner uniformement une suite maximisante, et qui rend possible l utilisation de banach alaoglu pour l existence

Corollary 3.2. *Let (u, v) be a pair of functions in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$. Then, there exists another pair (f, g) in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$ such that*

$$\mathsf{D}(u, v) \leq \mathsf{D}(f, g), \text{ and } \|f\|_\infty, \|g\|_\infty \leq M\|\mathsf{c}\|_\infty, \tag{52}$$

where D is in (10) and $M = \max(2, 3\frac{\beta+1}{\alpha+\beta})$.

juste ici il faut refaire la preuve. Ca change rien juste j'ai ecrit $\phi(u)$ donc faut remplacer u par v et je crois que quand on corrige cette confusion dans toute la suite ça fonctionne sans changer le résultat.

Proof: Let (u, v) be a pair of functions in $\mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$. From Lemma 3.4, the function $\Phi(u)$ satisfies $\operatorname{osc}(\Phi(u)) \leq 2\|\mathsf{c}\|_\infty$, where Φ is in (34). Thus, there exists a constant $A \in \mathbb{R}$ such that $g \triangleq \Phi(u) + A$ satisfies

$$\|g\|_\infty \leq 2\|\mathsf{c}\|_\infty. \tag{53}$$

From Lemma 3.2, it holds that $\mathsf{D}(u, \Phi(u)) = \mathsf{D}(u, g)$, so it follows from Corollary 3.1 that

$$\mathsf{D}(u, v) \leq \mathsf{D}(u, \phi(u)) = \mathsf{D}(u, g). \tag{54}$$

Now let $f \triangleq \Psi(g)$, where Ψ is in (36). From Corollary 3.1, the function f satisfies

$$\mathsf{D}(u, g) \leq \mathsf{D}(f, g). \tag{55}$$

Moreover, from Lemma 3.5, f also satisfies

$$\|f\|_\infty \leq \frac{\beta+1}{\alpha+\beta}(\|g\|_\infty + \|c\|_\infty). \tag{56}$$

Hence, combining the previous inequality with (53) yields

$$\|f\|_\infty \leq 3\frac{\beta+1}{\alpha+\beta}\|c\|_\infty, \tag{57}$$

The combination of (53) and (57) ensures that $\|f\|_\infty, \|g\|_\infty \leq M\|\mathsf{c}\|_\infty$ with $M = \max(2, 3\frac{\beta+1}{\alpha+\beta})$. This results in a pair (f,g) satisfying $\mathsf{D}(u,v) \leq \mathsf{D}(f,g)$, from (54) and (55), which concludes the proof. ■

Theorem 3.1. *The supremum in problem* (10) *is attained.*

Proof: Let $(u_n) \subset \mathrm{L}^\infty(\mathcal{X})$ and $(v_n) \subset \mathrm{L}^\infty(\mathcal{Y})$ be maximizing sequences. It follows from Corollary 3.2 that (u_n) and (v_n) can be chosen such that for all n, it holds that $\|u_n\|_\infty, \|v_n\|_\infty \leq M\|c\|_\infty$, with $M = \max(2, 3\frac{\beta+1}{\alpha+\beta})$. Application of the Banach-Alaoglu theorem [6, Theorem 3.16] yields existence of subsequences $(u_{n_k})_k$ and $(v_{n_k})_k$ such that

$$u_n \rightharpoonup^* \overline{u} \text{ in } \mathrm{L}^\infty(\mathcal{X}), \text{ and } v_n \rightharpoonup^* \overline{v} \text{ in } \mathrm{L}^\infty(\mathcal{Y}). \tag{58}$$

In particular, the following weak$*$ convergence also holds in $\mathrm{L}^\infty(\mathcal{X}\times\mathcal{Y})$

$$u_n + v_n - \mathsf{c} \rightharpoonup^* \overline{u} + \overline{v} - \mathsf{c}. \tag{59}$$

Thus, it follows from the convexity of exp and the weak$*$ lower semi-continuity of the integral functionals in the dual objective in (10) ([6, Theorem 3.7]) that

$$\liminf_{k\to\infty} \int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} \mathrm{d}Q(y) \geq \int_{\mathcal{Y}} e^{-\frac{\overline{v}(y)}{\beta}} \mathrm{d}Q(y); \text{ and} \tag{60}$$

$$\liminf_{k\to\infty} \int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y) \geq \int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\overline{u}(x)+\overline{v}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y). \tag{61}$$

From the continuity and monotonicity of log on $(0,\infty)$ and the two previous inequalities, it follows that

$$\liminf_{k\to\infty} \log\left(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}} \mathrm{d}Q(y)\right) \geq \log\left(\int_{\mathcal{Y}} e^{-\frac{\overline{v}(y)}{\beta}} \mathrm{d}Q(y)\right); \text{ and} \tag{62}$$

$$\liminf_{k\to\infty} \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y)\right) \geq \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\overline{u}(x)+\overline{v}(y)-\mathsf{c}(x,y)}{\alpha}} \mathrm{d}PQ(x,y)\right). \tag{63}$$

From which it can be deduced that

$$\begin{aligned}
&\sup_{u\in \mathrm{L}^\infty(\mathcal{X}),v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v)\\
&= \lim_{k\to\infty}\Bigg[\int_{\mathcal{X}} u_{n_k}(x)\mathrm{d}P(x) - \beta\log\Bigg(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}}\mathrm{d}Q(y)\Bigg)\\
&\quad -\alpha\log\Bigg(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\Bigg)\Bigg] \qquad (64)\\
&\leq \lim_{k\to\infty}\left[\int_{\mathcal{X}} u_{n_k}(x)\mathrm{d}P(x)\right] - \beta\liminf_{k\to\infty}\left[\log\left(\int_{\mathcal{Y}} e^{-\frac{v_{n_k}(y)}{\beta}}\mathrm{d}Q(y)\right)\right]\\
&\quad -\alpha\liminf_{k\to\infty}\left[\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{u_{n_k}(x)+v_{n_k}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right)\right]. \qquad (65)
\end{aligned}$$

It follows from the weak$*$ convergence of (u_{n_k}) to $\overline{u}$ that

$$\lim_{k\to\infty}\left[\int_{\mathcal{X}} u_{n_k}(x)\mathrm{d}P(x)\right] = \int_{\mathcal{X}} \overline{u}(x)\mathrm{d}P(x). \qquad (66)$$

Combining (62), (63), (65) and the previous equality yields

$$\begin{aligned}
\sup_{u\in \mathrm{L}^\infty(\mathcal{X}),v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \leq &\int_{\mathcal{X}} \overline{u}(x)\mathrm{d}P(x) - \beta\log\left(\int_{\mathcal{Y}} e^{-\frac{\overline{v}(y)}{\beta}}\mathrm{d}Q(y)\right)\\
&-\alpha\log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\overline{u}(x)+\overline{v}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right), \qquad (67)
\end{aligned}$$

or equivalently $\sup_{u\in \mathrm{L}^\infty(\mathcal{X}),v\in \mathrm{L}^\infty(\mathcal{Y})} \mathsf{D}(u,v) \leq \mathsf{D}(\overline{u},\overline{v})$, which concludes the proof. ■

3.3 Primal attainment

ici dire que parmi toutes les solutions au dual, on a fait exprès avec phi et psi de choisir celles qui vérifient naturellement des conditions de normalisation qui permettent d'écrire plus simplement la solution du probleme primal.

Theorem 3.2. *The infimum in problem* (5) *is attained at a unique measure* $\hat{\pi} \in \Pi(P,\star)$. *Moreover,* $\hat{\pi}$ *is absolutely continuous with respect to the product measure* PQ, *and its density can be written as*

$$\frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}(x,y) = e^{\frac{\hat{u}(x)}{\alpha}} e^{-\frac{\mathsf{c}(x,y)}{\alpha}} e^{\frac{\hat{v}(y)}{\alpha}}, \qquad (68)$$

where $(\hat{u},\hat{v})$ *is any pair of maximizing potentials for problem* (10) *satisfying* (33) *and* (35).

Proof: Let $(\hat{u}, \hat{v}) \in \mathrm{L}^\infty(\mathcal{X}) \times \mathrm{L}^\infty(\mathcal{Y})$ be a pair of maximizing potentials for problem (10) satisfying (33) and (35). Define $\hat{\pi}$ as in (68). Let $\hat{\rho}$ denote the density $\frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}$ and $\hat{s}$ denote the density $\frac{\mathrm{d}\hat{\pi}^{\mathcal{Y}}_{\#}}{\mathrm{d}Q}$. Condition (33) ensures that for all $x \in \mathcal{X}$, it holds that

$$\int_{\mathcal{Y}} \hat{\rho}(x,y)\mathrm{d}Q(y) = 1. \tag{69}$$

Thus, for any integrable function $f \in \mathrm{L}^1(\mathcal{X})$, Fubini's theorem [4, Theorem 8.8] yields

$$\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\hat{\pi}(x,y) = \int_{\mathcal{X}} f(x)\left(\int_{\mathcal{Y}} \hat{\rho}(x,y)\mathrm{d}Q(y)\right)\mathrm{d}P(x) = \int_{\mathcal{X}} f(x)\mathrm{d}P(x). \tag{70}$$

Hence, the measure $\hat{\pi}$ is a probability measure with first marginal $\hat{\pi}^{\mathcal{X}}_{\#} = P$, which implies $\hat{\pi} \in \Pi(P, \star)$. Besides, the optimality condition (35) implies

$$\int_{\mathcal{Y}} e^{-\hat{v}(y)/\beta}\mathrm{d}Q(y) = 1, \text{ and } \hat{s} = e^{-\hat{v}/\beta}. \tag{71}$$

From (69) and (71), the dual objective $\mathsf{D}(\hat{u}, \hat{v})$, with D in (10), can be computed as follows

$$\begin{aligned}\mathsf{D}(\hat{u}, \hat{v}) &= \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x) - \beta \log\left(\int_{\mathcal{Y}} e^{-\frac{\hat{v}(y)}{\beta}}\mathrm{d}Q(y)\right) \\ &\quad -\alpha \log\left(\int_{\mathcal{X}\times\mathcal{Y}} e^{\frac{\hat{u}(x)+\hat{v}(y)-\mathsf{c}(x,y)}{\alpha}}\mathrm{d}PQ(x,y)\right) \qquad (72)\\ &= \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x) - \beta\log(1) - \alpha\log(1) = \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x). \qquad (73)\end{aligned}$$

From (5), the primal objective value at $\hat{\pi}$ is defined via

$$\begin{aligned}\mathsf{P}(\hat{\pi}) = &\int_{\mathcal{X}\times\mathcal{Y}} (\mathsf{c}(x,y)\hat{\rho}(x,y) + \alpha(\hat{\rho}(x,y)\log(\hat{\rho}(x,y)) - \hat{\rho}(x,y) + 1))\mathrm{d}PQ(x,y) \\ &+\beta\int_{\mathcal{Y}} (\hat{s}(y)\log(\hat{s}(y)) - \hat{s}(y) + 1)\mathrm{d}Q(y). \qquad (74)\end{aligned}$$

For all points $(x,y) \in \mathcal{X} \times \mathcal{Y}$, since $\hat{\rho}(x,y) = e^{\frac{\hat{u}(x)+\hat{v}(y)-\mathsf{c}(x,y)}{\alpha}}$, the pointwise Fenchel-Young equality [1, Theorem 2.3.1] for the function $t \mapsto \alpha(t\log(t) - t + 1)$ states

$$\alpha(\hat{\rho}(x,y)\log(\hat{\rho}(x,y)) - \hat{\rho}(x,y) + 1) = \hat{\rho}(x,y)(\hat{u}(x) + \hat{v}(y) - \mathsf{c}(x,y)) - \alpha(\hat{\rho}(x,y) - 1). \tag{75}$$

Substitution into (74) leads to

$$\begin{aligned}\mathsf{P}(\hat{\pi}) = &\int_{\mathcal{X}\times\mathcal{Y}} \hat{\rho}(x,y)(\hat{u}(x) + \hat{v}(y))\mathrm{d}PQ(x,y) - \int_{\mathcal{X}\times\mathcal{Y}} \alpha(\hat{\rho}(x,y) - 1)\mathrm{d}PQ(x,y) \\ &+\beta\int_{\mathcal{Y}} (\hat{s}(y)\log(\hat{s}(y)) - \hat{s}(y) + 1)\mathrm{d}Q(y). \qquad (76)\end{aligned}$$

As $\hat{\rho}$ is a probability density, the intergral term $\int_{\mathcal{X}\times\mathcal{Y}}(\hat{\rho}(x,y)-1)\mathrm{d}PQ(x,y)$ vanishes. Equation (71) provides the relation $\log(\hat{s}) = -\hat{v}/\beta$. Pointwise Fenchel-Young equality [1, Theorem 2.3.1] for the second entropic term in (74) implies

$$\beta(\hat{s}(y)\log(\hat{s}(y)) - \hat{s}(y) + 1) = -\hat{s}(y)\hat{v}(y) - \beta(\hat{s}(y) - 1). \tag{77}$$

Linearity of the integral and Fubini's theorem [4, Theorem 8.8] yield

$$\begin{aligned}\mathsf{P}(\hat{\pi}) = &\int_{\mathcal{X}} \hat{u}(x)\left(\int_{\mathcal{Y}} \hat{\rho}(x,y)\mathrm{d}Q(y)\right)\mathrm{d}P(x) + \int_{\mathcal{Y}} \hat{v}(y)\left(\int_{\mathcal{X}} \hat{\rho}(x,y)\mathrm{d}P(x)\right)\mathrm{d}Q(y)\\ &- \int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y)\mathrm{d}Q(y) - \beta\int_{\mathcal{Y}}(\hat{s}(y)-1)\mathrm{d}Q(y). \end{aligned}\tag{78}$$

Definitions of the marginals $\hat{\pi}^{\mathcal{X}}_{\#}$ and $\hat{\pi}^{\mathcal{Y}}_{\#}$ coupled with identities (69) and (71) lead to

$$\mathsf{P}(\hat{\pi}) = \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x) + \int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y)\mathrm{d}Q(y) - \int_{\mathcal{Y}} \hat{v}(y)\hat{s}(y)\mathrm{d}Q(y) - \beta(1-1) \tag{79}$$

$$= \int_{\mathcal{X}} \hat{u}(x)\mathrm{d}P(x) = D(\hat{u},\hat{v}), \tag{80}$$

where the last equality follows from (73). Therefore, from Lemma 3.1, strong duality $\mathsf{P}(\hat{\pi}) = \mathsf{D}(\hat{u},\hat{v})$ holds and $\hat{\pi}$ is a minimizer for P. Uniqueness of the minimizer $\hat{\pi}$ follows from the strict convexity of the Kullback-Leibler divergence on its support. ■

faire une remarque que l'unicité du minimiseur primal permet en fait de justifier l'unicité du maximiseur dual qui vérifie simultanément (33) et (35), et que c'est ce qu'on fait dans le corollaire suivant.

Corollary 3.3. *There exists a unique pair of functions* $(u,v) \in \mathrm{L}^\infty(\mathcal{X})\times\mathrm{L}^\infty(\mathcal{Y})$ *satisfying* (33) *and* (35) *simultaneously.*

Proof: Let (u_1,v_1) and (u_2,v_2) satisfy (33) and (35) simultaneously, that is

$$u_1 = \Phi(v_1), \qquad v_1 = \Psi(u_1), \tag{81}$$

$$u_2 = \Phi(v_2), \qquad v_2 = \Psi(u_2), \tag{82}$$

Associate to each pair the coupling defined as follows

$$\pi_1(\mathrm{d}x,\mathrm{d}y) \triangleq \exp\left(\frac{u_1(x)+v_1(y)-\mathsf{c}(x,y)}{\alpha}\right) P(\mathrm{d}x)\,Q(\mathrm{d}y), \tag{83}$$

$$\pi_2(\mathrm{d}x,\mathrm{d}y) \triangleq \exp\left(\frac{u_2(x)+v_2(y)-\mathsf{c}(x,y)}{\alpha}\right) P(\mathrm{d}x)\,Q(\mathrm{d}y). \tag{84}$$

From Theorem 3.2, (81) and (82), π_1 and π_2 are primal minimizers. Uniqueness of the primal minimizer (Theorem 3.2) yields

$$\pi_1 = \pi_2, \tag{85}$$

which, together with (83) and (84), implies

$$\exp\left(\frac{u_1(x)+v_1(y)-\mathsf{c}(x,y)}{\alpha}\right) \overset{a.s.}{\underset{PQ}{=}} \exp\left(\frac{u_2(x)+v_2(y)-\mathsf{c}(x,y)}{\alpha}\right). \tag{86}$$

Taking logarithms in the previous equality gives

$$u_1(x)+v_1(y) \overset{a.s.}{\underset{PQ}{=}} u_2(x)+v_2(y). \tag{87}$$

Fix y_0 outside a Q-negligible set such that (87) holds for P-a.e. x. Then, for P-a.e. x,

$$u_1(x)-u_2(x)=v_2(y_0)-v_1(y_0) \overset{a.s.}{\underset{P}{=}} K, \tag{88}$$

for the constant $K \triangleq v_2(y_0)-v_1(y_0)$. Insert (88) into (87) to obtain, for Q-a.e. y,

$$v_2(y)-v_1(y)=K. \tag{89}$$

Finally, combining (89) with the normalization (48), which follows from (81) and (82), yields

$$1=\int_{\mathcal{Y}} \exp\left(-\frac{v_2(y)}{\beta}\right) \mathrm{d}Q(y) \tag{90}$$

$$=\int_{\mathcal{Y}} \exp\left(-\frac{v_1(y)+K}{\beta}\right) \mathrm{d}Q(y) \tag{91}$$

$$=e^{-K/\beta}\int_{\mathcal{Y}} \exp\left(-\frac{v_1(y)}{\beta}\right) \mathrm{d}Q(y)=e^{-K/\beta}, \tag{92}$$

which implies $K=0$, hence (88) and (89) yield

$$u_1 \overset{a.s.}{\underset{P}{=}} u_2, \qquad v_1 \overset{a.s.}{\underset{Q}{=}} v_2. \tag{93}$$

This proves uniqueness of the pair satisfying (33) and (35). ■

3.4 (α,β)-approximation error control renommer la section c naze

Ici il faut faire l'intro, certainement citer les résultats de ceux qui ont travaillé sur des problèmes régularisés en OT en général auparavant. En gros on veut regarder comment le minimiseur du problème régularisé approxime le minimiseur du vrai problème quand α et β varient. Et pareil pour la valeeur obtenue aussi. Dans cette partie il faut bien mettre l'accent sur un détail important: à cause de la présence de "l'a priori" Q, la 2e marginale de la solution du pb régularisé est contrainte à avoir le même support que Q. Donc déjà on ne pourra pas converger vers un minimiseur du vrai pb sauf s'il y en a un qui est absolument continu par rapport à Q. C'est quand même a bien garder en tête parce que ça donne une bonne idée de comment choisir Q.

3.5 Iterative algorithms

3.5.1 Presentation of the algorithms

la il faut faire une intro bien solide. Dire donc que comme le sinkhorn classique l'idée repose simplement sur de la maximisation alternée. Pourquoi on choisit les itérations avec Φ et Ψ particulierement ? Parce que parmi les potentiels maximiseurs, Φ et Ψ ont été construits pour donner ceux qui vérifient naturellement une contrainte de normalisation, pour que la mesure associée soit direct une proba sans avoir besoin de normaliser. Si on imposait pas ça, a chaque itération il y aurait des intégrales au dénominateur. Clairement c'est a comparer avec la

Algorithm 1: Unscaled iterative procedure

Let $v^{(0)}$ be in $\mathrm{L}^\infty(\mathcal{Y})$.

$$u^{(l+1)} = \Phi(v^{(l)}); \tag{94a}$$
$$v^{(l+1)} = \Psi(u^{(l+1)}), \tag{94b}$$

where Φ is in (34) and Ψ is in (36).

procédure proposée par Chizat dans [3, Equation 3.6]. EN fait on voit que ce qu'on aurait obtenu c'est exactement la meme chose sauf que notre opérateur Ψ a un terme en plus (la constante qui est le premier terme dans (36)), qui en plus est certainement chiant a calculer mais offre une meilleure stabiloté au moins sur le plan théorique (resultats de convergence bien meilleurs) et certainement aussi pour l'implémentation

3.5.2 Theoretical guarantees

In the more classical unbalanced Sinkhorn setting with two relaxed marginals [3, Theorem 3.11], a linear rate of convergence can be obtained by viewing each scaling update as an order-preserving map that is z-homogeneous (with $0 < z < 1$) and therefore strictly contractive in the Thompson metric [3, Definition 3.10]. In this semi-relaxed setting, however, the scaled updates exhibit a global renormalization in (36) that destroys the required homogeneity structure in Thompson's geometry. More precisely, with the Gibbs kernel $e^{-c(x,y)/\alpha}$, the first update in (95a) is the analogue of the hard marginal projection: it is order-reversing and (-1)-homogeneous, hence non-expansive (in fact 1-Lipschitz) in projective metrics. The second update in (95b) involves a normalization factor that depends on the current iterate but is constant with respect to the spatial variable. This global rescaling is precisely what prevents the update from being z-homogeneous with $z \in (0, 1)$ in the sense required by the Thompson contraction argument. By contrast, the Hilbert projective metric is invariant under multiplication by positive scalars, so such global renormalizations are invisible at the level of distances [7, Lemma 2.1(iv)]. This makes the Hilbert geometry the natural framework for the scaled scheme.

Algorithm 2: Scaled iterative procedure

Let $b^{(0)} \in \mathrm{L}^\infty(\mathcal{Y})$ be equal to 1, and for all $l \in \mathbb{N}$, let the following terms of the sequence be defined as

$$a^{(l+1)} = \Xi\left(b^{(l)}\right); \tag{95a}$$

$$v^{(l+1)} = \Gamma\left(a^{(l+1)}\right), \tag{95b}$$

where the operators $\Xi : \mathrm{L}^\infty(\mathcal{Y}) \to \mathrm{L}^\infty(\mathcal{X})$ and $\Gamma : \mathrm{L}^\infty(\mathcal{X}) \to \mathrm{L}^\infty(\mathcal{Y})$ are defined by

$$\Xi(b)(x) \triangleq \frac{1}{\int_{\mathcal{Y}} b(y) e^{-\frac{\mathsf{c}(x,y)}{\alpha}} \mathrm{d}Q(y)}; \tag{96}$$

$$\Gamma(a)(y) \triangleq \left(\frac{1}{\int_{\mathcal{X}} a(x) e^{-\frac{\mathsf{c}(x,y)}{\alpha}} \mathrm{d}P(x)}\right)^{\frac{\beta}{\alpha+\beta}} \times \left(\int_{\mathcal{Y}} \left(\int_{\mathcal{X}} a(x) e^{\frac{-\mathsf{c}(x,y')}{\alpha}} \mathrm{d}P(x)\right)^{\frac{\alpha}{\alpha+\beta}} \mathrm{d}Q(y')\right)^{\frac{\beta}{\alpha}}. \tag{97}$$

Definition 3.1 (Hilbert metric)**.** *Let $\mathrm{L}^\infty_{++}(\mathcal{X})$ be the subset of $\mathrm{L}^\infty(\mathcal{X})$ defined by*

$$\mathrm{L}^\infty_{++}(\mathcal{X}) \triangleq \{a \in \mathrm{L}^\infty(\mathcal{X}) : \inf_{x\in\mathcal{X}} a(x) > 0\} \tag{98}$$

and similarly for $\mathrm{L}^\infty_{++}(\mathcal{Y})$. Given two functions $f, g \in \mathrm{L}^\infty_{++}(\mathcal{Y})$, Hilbert's projective metric is given by

$$d_{\mathrm{H}}(f,g) \triangleq \log\left(\frac{M(f/g)}{m(f/g)}\right), \tag{99}$$

$$\text{where } M(f/g) \triangleq \sup_{y\in\mathcal{Y}} \frac{f(y)}{g(y)}, \text{ and } m(f/g) \triangleq \inf_{y\in\mathcal{Y}} \frac{f(y)}{g(y)}, \tag{100}$$

which is exactly Birkhoff's version of Hilbert's metric on cones [7, Equation (2.2)].

Theorem 3.3. *Let $\hat{u}$ and $\hat{v}$ be the unique maximizers for problem* (10) *that satisfy* (33) *and* (35) *simultaneously, and define the associated scaled potentials*

$$\hat{a} = \exp\left(\frac{\hat{u}}{\alpha}\right), \qquad \hat{b} = \exp\left(\frac{\hat{v}}{\alpha}\right). \tag{101}$$

Then, the scaled iterates defined in Algorithm 2 converge to these scaled potentials at a linear rate: for all $l \geq 0$,

$$d_{\mathrm{H}}\left(b^{(l)}, \hat{b}\right) \leq \rho^l_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{102}$$

$$d_{\mathrm{H}}\left(a^{(l+1)}, \hat{a}\right) \leq \tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha}\right) \rho^l_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{103}$$

where d_{H} is the Hilbert metric in Definition 3.1 and

$$\rho_{(\mathsf{c},\alpha,\beta)} \triangleq \frac{\beta}{\alpha+\beta}\tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha}\right)^2. \tag{104}$$

la preuve suivante je l'ai bcp refaite donc a bien checker. Il faudra dire que c des trucs standards issus de méthodes pour les sinkhorn de base. Et aussi préciser un peu tout le vcabulaire technique sur les order preserving et homogenous of degree 1. Peut etre aussi pour les part of the cone meme si c connu normalement. Il manque aussi des from blablabla et des from blablabla pour les liens entre les inégalités.

Proof: The proof is organized as follows. First, two positive linear operators are introduced and their projective diameters shown to be are bounded. Theorem 2.9 in [7] then yields strict contractions in Hilbert's metric. Next, the operators Ξ and Γ are shown to be Lipschitz, and the one-step map $\Gamma\circ\Xi$ is shown to be a strict contraction. Finally, the normalization built into Γ identifies a complete slice of the positive cone, and the Banach fixed-point theorem yields geometric convergence of the scaled iterates. The identification of the limit with the normalized maximizing potentials concludes the argument. Define the positive linear operators

$$\mathrm{K} : \mathrm{L}^\infty(\mathcal{Y}) \to \mathrm{L}^\infty(\mathcal{X}), \qquad (\mathrm{K}\,b)(x) \triangleq \int_{\mathcal{Y}} e^{-\frac{\mathsf{c}(x,y)}{\alpha}}\, b(y)\,\mathrm{d}Q(y), \tag{105}$$

$$\mathrm{K}^\mathsf{T} : \mathrm{L}^\infty(\mathcal{X}) \to \mathrm{L}^\infty(\mathcal{Y}), \qquad (\mathrm{K}^\mathsf{T}\,a)(y) \triangleq \int_{\mathcal{X}} e^{-\frac{\mathsf{c}(x,y)}{\alpha}}\, a(x)\,\mathrm{d}P(x). \tag{106}$$

Since c is bounded, it holds for all $(x,y)\in\mathcal{X}\times\mathcal{Y}$ that

$$e^{-\frac{\|\mathsf{c}\|_\infty}{\alpha}} \;\le\; e^{-\frac{\mathsf{c}(x,y)}{\alpha}} \;\le\; e^{\frac{\|\mathsf{c}\|_\infty}{\alpha}}. \tag{107}$$

Hence, for every $b\in\mathrm{L}^\infty_+(\mathcal{Y})\setminus\{0\}$ and every $x\in\mathcal{X}$,

$$e^{-\frac{\|\mathsf{c}\|_\infty}{\alpha}}\int_{\mathcal{Y}} b(y)\,\mathrm{d}Q(y) \;\le\; (\mathrm{K}\,b)(x) \;\le\; e^{\frac{\|\mathsf{c}\|_\infty}{\alpha}}\int_{\mathcal{Y}} b(y)\,\mathrm{d}Q(y). \tag{108}$$

Since $\int_{\mathcal{Y}} b(y)\,\mathrm{d}Q(y) > 0$, this yields $\mathrm{K}\,b\in\mathrm{L}^\infty_{++}(\mathcal{X})$. The same argument gives $\mathrm{K}^\mathsf{T}\,a\in\mathrm{L}^\infty_{++}(\mathcal{Y})$ for every $a\in\mathrm{L}^\infty_+(\mathcal{X})\setminus\{0\}$. If $b_1,b_2\in\mathrm{L}^\infty_+(\mathcal{Y})\setminus\{0\}$, then for every $x\in\mathcal{X}$,

$$\frac{(\mathrm{K}\,b_1)(x)}{(\mathrm{K}\,b_2)(x)} \le e^{\frac{2\|\mathsf{c}\|_\infty}{\alpha}}\frac{\int_{\mathcal{Y}} b_1(y)\,\mathrm{d}Q(y)}{\int_{\mathcal{Y}} b_2(y)\,\mathrm{d}Q(y)}, \tag{109}$$

$$\frac{(\mathrm{K}\,b_1)(x)}{(\mathrm{K}\,b_2)(x)} \ge e^{-\frac{2\|\mathsf{c}\|_\infty}{\alpha}}\frac{\int_{\mathcal{Y}} b_1(y)\,\mathrm{d}Q(y)}{\int_{\mathcal{Y}} b_2(y)\,\mathrm{d}Q(y)}. \tag{110}$$

Taking the essential supremum and the essential infimum over $x\in\mathcal{X}$ yields

$$d_{\mathrm{H}}(\mathrm{K}\,b_1,\mathrm{K}\,b_2) \;\le\; \frac{4\|\mathsf{c}\|_\infty}{\alpha}. \tag{111}$$

Hence the projective diameter of K is finite, with

$$\Delta(\mathrm{K}) \leq \frac{4\|\mathsf{c}\|_\infty}{\alpha}. \tag{112}$$

The same argument yields

$$\Delta(\mathrm{K}^\mathsf{T}) \leq \frac{4\|\mathsf{c}\|_\infty}{\alpha}. \tag{113}$$

Since $\Delta(\mathrm{K}) < +\infty$ and $\Delta(\mathrm{K}^\mathsf{T}) < +\infty$, Theorem 2.9 in [7] applies to the linear maps

$$\mathrm{K} : \mathrm{L}^\infty(\mathcal{Y}) \to \mathrm{L}^\infty(\mathcal{X}) \qquad \text{and} \qquad \mathrm{K}^\mathsf{T} : \mathrm{L}^\infty(\mathcal{X}) \to \mathrm{L}^\infty(\mathcal{Y}), \tag{114}$$

with source and target cones $\mathrm{L}^\infty_+(\mathcal{Y})$ and $\mathrm{L}^\infty_+(\mathcal{X})$. It follows that K is a strict contraction, for Hilbert's metric, from the part $\mathrm{L}^\infty_{++}(\mathcal{Y})$ of $\mathrm{L}^\infty_+(\mathcal{Y})$ into the part $\mathrm{L}^\infty_{++}(\mathcal{X})$ of $\mathrm{L}^\infty_+(\mathcal{X})$, and that K^T is a strict contraction, for Hilbert's metric, from $\mathrm{L}^\infty_{++}(\mathcal{X})$ into $\mathrm{L}^\infty_{++}(\mathcal{Y})$, with contraction factors $\tanh(\Delta(\mathrm{K})/4)$ and $\tanh(\Delta(\mathrm{K}^\mathsf{T})/4)$, respectively. Hence, for all $b_1, b_2 \in \mathrm{L}^\infty_{++}(\mathcal{Y})$ and all $a_1, a_2 \in \mathrm{L}^\infty_{++}(\mathcal{X})$, it holds that

$$d_\mathrm{H}(\mathrm{K}b_1,\mathrm{K}b_2) \leq \tanh\left(\frac{\Delta(\mathrm{K})}{4}\right) d_\mathrm{H}(b_1,b_2) \leq \tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha}\right) d_\mathrm{H}(b_1,b_2), \tag{115}$$

$$d_\mathrm{H}\Big(\mathrm{K}^\mathsf{T}a_1,\mathrm{K}^\mathsf{T}a_2\Big) \leq \tanh\left(\frac{\Delta(\mathrm{K}^\mathsf{T})}{4}\right) d_\mathrm{H}(a_1,a_2) \leq \tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha}\right) d_\mathrm{H}(a_1,a_2). \tag{116}$$

The first scaled update satisfies

$$\Xi(b)(x) = \frac{1}{(\mathrm{K}\, b)(x)}, \tag{117}$$

for every $b \in \mathrm{L}^\infty_{++}(\mathcal{Y})$ and every $x \in \mathcal{X}$. Moreover, relation (99) implies that, for every pair f, g in either $\mathrm{L}^\infty_{++}(\mathcal{X})$ or $\mathrm{L}^\infty_{++}(\mathcal{Y})$,

$$d_\mathrm{H}(f^{-1}, g^{-1}) = d_\mathrm{H}(f, g). \tag{118}$$

Consequently, for all $b_1, b_2 \in \mathrm{L}^\infty_{++}(\mathcal{Y})$,

$$\begin{aligned} d_\mathrm{H}(\Xi(b_1), \Xi(b_2)) &= d_\mathrm{H}\big((\mathrm{K}\, b_1)^{-1}, (\mathrm{K}\, b_2)^{-1}\big) && (119)\\ &= d_\mathrm{H}(\mathrm{K}\, b_1, \mathrm{K}\, b_2) && (120)\\ &\leq \tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha}\right) d_\mathrm{H}(b_1, b_2). && (121) \end{aligned}$$

Let $a \in \mathrm{L}^\infty_{++}(\mathcal{X})$. The second scaled update can be written as

$$\Gamma(a)(y) = \Big(\mathrm{K}^\mathsf{T}\, a(y)\Big)^{-\frac{\beta}{\alpha+\beta}} \left(\int_\mathcal{Y} \Big(\mathrm{K}^\mathsf{T}\, a(y')\Big)^{\frac{\alpha}{\alpha+\beta}} \, \mathrm{d}Q(y')\right)^{\frac{\beta}{\alpha}}, \tag{122}$$

and $\Gamma(a) \in \mathrm{L}^\infty_{++}(\mathcal{Y})$. The second factor is a positive scalar independent of y. Lemma 2.1(iv) in [7] therefore gives, for all $a_1, a_2 \in \mathrm{L}^\infty_{++}(\mathcal{X})$,

$$d_{\mathrm{H}}(\Gamma(a_1), \Gamma(a_2)) = d_{\mathrm{H}}\Big(\big(\mathrm{K}^\mathsf{T} a_1\big)^{-\frac{\beta}{\alpha+\beta}}, \big(\mathrm{K}^\mathsf{T} a_2\big)^{-\frac{\beta}{\alpha+\beta}} \Big). \tag{123}$$

The map $f \mapsto f^{\frac{\beta}{\alpha+\beta}}$ is order-preserving and homogeneous of degree $\frac{\beta}{\alpha+\beta}$. Proposition 2.6 in [7] then implies

$$d_{\mathrm{H}}\Big(f^{\frac{\beta}{\alpha+\beta}}, g^{\frac{\beta}{\alpha+\beta}} \Big) \;\leq\; \frac{\beta}{\alpha+\beta}\, d_{\mathrm{H}}(f, g) \tag{124}$$

for all $f, g \in \mathrm{L}^\infty_{++}(\mathcal{Y})$. Together with (118), this yields

$$d_{\mathrm{H}}(\Gamma(a_1), \Gamma(a_2)) \leq \frac{\beta}{\alpha+\beta}\, d_{\mathrm{H}}\Big(\mathrm{K}^\mathsf{T} a_1, \mathrm{K}^\mathsf{T} a_2 \Big) \tag{125}$$

$$\leq \frac{\beta}{\alpha+\beta} \tanh\Big(\frac{\|\mathsf{c}\|_\infty}{\alpha} \Big)\, d_{\mathrm{H}}(a_1, a_2). \tag{126}$$

Combining (121) and (126) shows that the one-step map $T \triangleq \Gamma \circ \Xi$ satisfies, for all $b_1, b_2 \in \mathrm{L}^\infty_{++}(\mathcal{Y})$,

$$d_{\mathrm{H}}(T(b_1), T(b_2)) \leq \frac{\beta}{\alpha+\beta} \tanh\Big(\frac{\|\mathsf{c}\|_\infty}{\alpha} \Big)^2\, d_{\mathrm{H}}(b_1, b_2). \tag{127}$$

It remains to identify a complete metric space left invariant under T. For every $a \in \mathrm{L}^\infty_{++}(\mathcal{X})$ and every $y \in \mathcal{Y}$,

$$\Gamma(a)(y)^{-\alpha/\beta} = \frac{\Big(\mathrm{K}^\mathsf{T} a(y)\Big)^{\frac{\alpha}{\alpha+\beta}}}{\int_{\mathcal{Y}} \Big(\mathrm{K}^\mathsf{T} a(y')\Big)^{\frac{\alpha}{\alpha+\beta}}\, \mathrm{d}Q(y')}. \tag{128}$$

Integration with respect to Q yields

$$\int_{\mathcal{Y}} \Gamma(a)(y)^{-\alpha/\beta}\, \mathrm{d}Q(y) = 1. \tag{129}$$

Hence every iterate $b^{(l)}$ belongs to

$$\Sigma \triangleq \Big\{ b \in \mathrm{L}^\infty_{++}(\mathcal{Y}) : \int_{\mathcal{Y}} b(y)^{-\alpha/\beta}\, \mathrm{d}Q(y) = 1 \Big\}. \tag{130}$$

Consider

$$q(b) \triangleq \Big(\int_{\mathcal{Y}} b(y)^{-\alpha/\beta}\, \mathrm{d}Q(y) \Big)^{-\beta/\alpha}. \tag{131}$$

This map is homogeneous of degree 1. It is also continuous on $\mathrm{L}^\infty_{++}(\mathcal{Y})$ for the L^∞ norm. Indeed, let b be in $\mathrm{L}^\infty_{++}(\mathcal{Y})$ and $(b_n)_{n\geq 1}$ be a sequence in $\mathrm{L}^\infty_{++}(\mathcal{Y})$ such that $b_n \to b$ in $\mathrm{L}^\infty(\mathcal{Y})$, and set

$$m_b \triangleq \operatorname*{ess\,inf}_Q b > 0. \tag{132}$$

There exists $n_0 \in \mathbb{N}$ such that $b_n(y) \geq m_b/2$ for Q-almost every $y \in \mathcal{Y}$ and every $n \geq n_0$. The map $t \mapsto t^{-\alpha/\beta}$ is Lipschitz on $[m_b/2, +\infty)$, hence

$$\left\| b_n^{-\alpha/\beta} - b^{-\alpha/\beta} \right\|_\infty \xrightarrow[n\to+\infty]{} 0. \tag{133}$$

Therefore

$$\int_{\mathcal{Y}} b_n(y)^{-\alpha/\beta} \, \mathrm{d}Q(y) \xrightarrow[n\to+\infty]{} \int_{\mathcal{Y}} b(y)^{-\alpha/\beta} \, \mathrm{d}Q(y), \tag{134}$$

and then $q(b_n) \to q(b)$. The cone $\mathrm{L}^\infty_+(\mathcal{Y})$ is closed and normal in the Banach space $\mathrm{L}^\infty(\mathcal{Y})$, and $\mathrm{L}^\infty_{++}(\mathcal{Y})$ is the part containing the constant function 1. Lemma 2.1 in [8], applied with the constant function 1, yields that (Σ, d_{H}) is a complete metric space. Since $b^{(0)} \equiv 1$ belongs to Σ and $T(\Sigma) \subset \Sigma$ follows from (129), the Banach fixed-point theorem implies the existence of a unique fixed point $\hat{b} \in \Sigma$ and gives, for all $l \geq 0$,

$$d_{\mathrm{H}}\left(b^{(l)}, \hat{b}\right) \leq \left(\frac{\beta}{\alpha+\beta} \tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha} \right)^2 \right)^l d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{135}$$

which is exactly (102). Define

$$\hat{a} \triangleq \Xi(\hat{b}). \tag{136}$$

Then, for every $l \geq 1$,

$$\begin{aligned} d_{\mathrm{H}}\left(a^{(l)}, \hat{a}\right) &= d_{\mathrm{H}}\left(\Xi(b^{(l-1)}), \Xi(\hat{b})\right) && (137)\\ &\leq \tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha} \right) d_{\mathrm{H}}\left(b^{(l-1)}, \hat{b}\right) && (138)\\ &\leq \tanh\left(\frac{\|\mathsf{c}\|_\infty}{\alpha} \right) \rho^{l-1}_{(\mathsf{c},\alpha,\beta)} \, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), && (139) \end{aligned}$$

which is exactly (103).

Finally, define

$$\hat{u} \triangleq \alpha \log(\hat{a}) \in \mathrm{L}^\infty(\mathcal{X}), \qquad \hat{v} \triangleq \alpha \log(\hat{b}) \in \mathrm{L}^\infty(\mathcal{Y}). \tag{140}$$

The fixed point relation $T(\hat{b}) = \hat{b}$ and the definition $\hat{a} = \Xi(\hat{b})$ are equivalent to

$$\hat{u} = \Phi(\hat{v}), \qquad \hat{v} = \Psi(\hat{u}), \tag{141}$$

where Φ and Ψ are defined in (34) and (36). Corollary 3.1 and Corollary 3.3 then imply that $\hat{u}$ and $\hat{v}$ are the unique maximizers for problem (10) that satisfy (33) and (35) simultaneously. Therefore, $\hat{a}$ and $\hat{b}$ are the associated unique scaled maximizing potentials. ■

A priori le + dur est fait maintenant l objectif est de deduire la convergence en objectif a une vitesse lineaire aussi, et pourquoi pas des convergences en

norme uniforme a vitesse lineaire pour les potentiels (mais alors peut etre faut il se restreindre au cas ou X et Y sont finis, c'est sur que c deja traite dans la littérature pour le coup). Dans tous les cas aussi la convergence de la version "primale" des itérés doit etre traitée rapidement aussi. Après ça je ne vois que l implementation qui reste.

Corollary 3.4. *The non-scaled iterates* $(u^{(l)}, v^{(l)})$ *defined in Algorithm 1 converge uniformly to* $(\hat{u}, \hat{v})$*, the unique pair of maximizers for problem* (10) *that satisfy* (33) *and* (35) *simultaneously, at a linear rate: for all* $l \geq 0$*,*

$$\|v^{(l)} - \hat{v}\|_\infty \leq \alpha \rho^l_{(\mathsf{c},\alpha,\beta)} d_{\mathrm{H}}\Big(b^{(0)}, \hat{b}\Big), \tag{142}$$

$$\|u^{(l+1)} - \hat{u}\|_\infty \leq \alpha \rho^l_{(\mathsf{c},\alpha,\beta)} d_{\mathrm{H}}\Big(b^{(0)}, \hat{b}\Big). \tag{143}$$

Proof: The proof is split into three parts; the first one deals with (142), the second one with (143), and the third one with (**??**). Let $(u^{(l)}, v^{(l)})$ be the unscaled iterates defined by Algorithm (1), $(a^{(l)}, b^{(l)})$ be the scaled iterates defined by Algorithm (2), and $(\hat{u}, \hat{v})$ be the unique pair of maximizers satisfying (33) and (35) simultaneously. Recall that the two parametrizations are linked through

$$u^{(l)} = \alpha \log\big(a^{(l)}\big), \qquad v^{(l)} = \alpha \log\big(b^{(l)}\big). \tag{144}$$

Hence, it holds that

$$\|v^{(l)} - \hat{v}\|_\infty = \alpha \left\| \log\left(\frac{b^{(l)}}{\hat{b}}\right) \right\|_\infty. \tag{145}$$

Define the pointwise ratio

$$r^{(l)}(y) \triangleq \frac{b^{(l)}(y)}{\hat{b}(y)}, \qquad y \in \mathcal{Y}, \tag{146}$$

and its bounds

$$m^{(l)} \triangleq \inf_{y \in \mathcal{Y}} r^{(l)}(y), \qquad M^{(l)} \triangleq \sup_{y \in \mathcal{Y}} r^{(l)}(y). \tag{147}$$

By definition of Hilbert's projective metric on $\mathrm{L}^\infty_{++}(\mathcal{Y})$ (Definition 3.1), it holds that

$$d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big) = \log\left(\frac{M^{(l)}}{m^{(l)}}\right). \tag{148}$$

It remains to show that

$$m^{(l)} \leq 1 \leq M^{(l)}. \tag{149}$$

Since the iterates $b^{(l)}$ are in Σ for all l and $\hat{b} \in \Sigma$, where Σ is in (130), the same gauge constraint as in (129) holds

$$\int_{\mathcal{Y}} \big(b^{(l)}(y)\big)^{-\alpha/\beta} \, dQ(y) = 1, \qquad \text{and} \quad \int_{\mathcal{Y}} \big(\hat{b}(y)\big)^{-\alpha/\beta} \, dQ(y) = 1. \tag{150}$$

The map $t \mapsto t^{-\alpha/\beta}$ is strictly decreasing on $(0, +\infty)$. Assume by contradiction that $M^{(l)} < 1$. Then $r^{(l)}(y) < 1$ for Q-a.e. y, i.e., $b^{(l)}(y) < \hat{b}(y)$ for Q-a.e. y, and the strict monotonicity of the integral implies

$$\int_{\mathcal{Y}} \big(b^{(l)}(y)\big)^{-\alpha/\beta}\, dQ(y) > \int_{\mathcal{Y}} \big(\hat{b}(y)\big)^{-\alpha/\beta}\, dQ(y), \tag{151}$$

which contradicts (150). Therefore $M^{(l)} \geq 1$. The case $m^{(l)} > 1$ is ruled out similarly. Thus, the inequalities in (149) are established. Combining (149) with $\|\log r^{(l)}\|_\infty = \max\{-\log m^{(l)}, \log M^{(l)}\}$ gives

$$\left\|\log\left(\frac{b^{(l)}}{\hat{b}}\right)\right\|_\infty = \max\{-\log m^{(l)}, \log M^{(l)}\} \tag{152}$$

$$\leq \log\left(\frac{M^{(l)}}{m^{(l)}}\right) \tag{153}$$

$$= d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big), \tag{154}$$

where the last equality follows from (148). Plugging (154) into (145) yields

$$\|v^{(l)} - \hat{v}\|_\infty \leq \alpha\, d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big). \tag{155}$$

Combining the previous inequality with (102) concludes the proof of (142). Regarding the proof of (143), recall from Algorithm 1 that

$$u^{(l+1)} = \Phi\big(v^{(l)}\big), \qquad \text{and } \hat{u} = \Phi(\hat{v}), \tag{156}$$

where Φ is in (34). Let v, w be in $\mathrm{L}^\infty(\mathcal{Y})$ and set $\delta \triangleq \|v - w\|_\infty$. Then, for Q-a.e. y in $\mathcal{Y}$, it holds that

$$-\delta \leq v(y) - w(y) \leq \delta \quad \Longrightarrow \quad e^{-\delta/\alpha} \leq \exp\left(\frac{v(y) - w(y)}{\alpha}\right) \leq e^{\delta/\alpha}. \tag{157}$$

Multiplying by $\exp\left(\frac{w(y) - \mathsf{c}(x,y)}{\alpha}\right)$ and integrating with respect to Q yields, for every $x \in \mathcal{X}$,

$$e^{-\delta/\alpha} \int_{\mathcal{Y}} \exp\left(\frac{w(y) - \mathsf{c}(x,y)}{\alpha}\right) dQ(y) \leq \int_{\mathcal{Y}} \exp\left(\frac{v(y) - \mathsf{c}(x,y)}{\alpha}\right) dQ(y) \tag{158}$$

$$\leq e^{\delta/\alpha} \int_{\mathcal{Y}} \exp\left(\frac{w(y) - \mathsf{c}(x,y)}{\alpha}\right) dQ(y). \tag{159}$$

Applying $-\alpha$ times the logarithm to the previous inequalities yields, for every $x \in \mathcal{X}$,

$$\Phi(w)(x) - \delta \leq \Phi(v)(x) \leq \Phi(w)(x) + \delta, \tag{160}$$

hence

$$\|\Phi(v) - \Phi(w)\|_\infty \leq \|v - w\|_\infty. \tag{161}$$

Combining (156) with (161) yields

$$\|u^{(l+1)} - \hat{u}\|_\infty = \|\Phi(v^{(l)}) - \Phi(\hat{v})\|_\infty \leq \|v^{(l)} - \hat{v}\|_\infty. \tag{162}$$

Together with (142), this implies

$$\|u^{(l+1)} - \hat{u}\|_\infty \leq \alpha\, d_{\mathrm{H}}\big(b^{(l)}, \hat{b}\big). \tag{163}$$

Finally, applying (102) concludes the proof of (143). ■

Que reste-t-il à faire ? La convergence en objectif, la convergence des mesures associées par le primal sont des trucs qu'on va s'empresser d'obtenir. A FAIRE EN PRIORITE. On a toutes les clés en main faut juste le faire. Ensuite il reste l'immense partie expérimentale à traiter, chose qu'il faudra faire très sérieusement, en planifiant très très bien chaque étape avant de commencer, et surtout SANS s'éparpiller. Peut être même dans un autre rapport en fait ? En tout cas le plus gros questionnement reste le controle en α, β de l'erreur faite par rapport au vrai problème non régularisé. Peut-être qu'il faudrait dans un permier temps se contenter de la Γ convergence qui peut apparemment être obtenue sans trop de gymnastique. Mais une borne satisfaisante sur cette erreur serait la cloture parfaite à tout ce dont on dispose... J'ai fait ca c bon

Corollary 3.5. *Let $(u^{(l)}, v^{(l)})_{l\geq 0}$ be the iterates defined by Algorithm 1. The sequence $\big(\mathsf{D}(u^{(l)}, v^{(l)})\big)_{l\geq 0}$ is non-decreasing and converges to $\mathsf{P}(\hat{\pi})$ at a linear speed:*

$$0 \;\leq\; \mathsf{P}(\hat{\pi}) - \mathsf{D}\Big(u^{(l)}, v^{(l)}\Big) \;\leq\; 4\alpha\, \rho^{l}_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\Big(b^{(0)}, \hat{b}\Big), \tag{164}$$

where $\rho_{(\mathsf{c},\alpha,\beta)}$ is defined in (104) *and $\hat{b} \triangleq e^{\hat{v}/\alpha}$.*

Proof: From Corollary 3.1, each alternating maximization step does not decrease the value of D, hence $\mathsf{D}(u^{(l)}, v^{(l)}) \leq \mathsf{D}(u^{(l+1)}, v^{(l+1)})$ for all $l \geq 0$. Strong duality, established in Theorem 3.2, provides the left-hand inequality in (164). The right-hand inequality rests on the following observation: for any measurable bounded functions f and g on a probability space (Ω, μ), it holds that

$$\left|\log\int_\Omega e^{f(t)}\,\mathrm{d}\mu(t) \;-\; \log\int_\Omega e^{g(t)}\,\mathrm{d}\mu(t)\right| \;\leq\; \|f-g\|_\infty. \tag{165}$$

Indeed, $\int e^{f(t)}\,\mathrm{d}\mu(t) \leq e^{\|f-g\|_\infty}\int e^{g(t)}\,\mathrm{d}\mu(t)$, from which it cqn be deduced that

$$\log\int_\Omega e^{f(t)}\,\mathrm{d}\mu(t) - \log\int_\Omega e^{g(t)}\,\mathrm{d}\mu(t) \leq \|f-g\|_\infty. \tag{166}$$

The reverse inequality is obtained symmetrically. From definition (10), the gap

decomposes as

$$\begin{aligned}\mathsf{D}(\hat{u},\hat{v})-\mathsf{D}(u^{(l)},v^{(l)}) = &\int_{\mathcal{X}}\big(\hat{u}-u^{(l)}\big)\,\mathrm{d}P \\ &-\beta\Bigg[\log\int_{\mathcal{Y}} e^{-\hat{v}(y)/\beta}\,\mathrm{d}Q(y) \;-\; \log\int_{\mathcal{Y}} e^{-v^{(l)}(y)/\beta}\,\mathrm{d}Q(y)\Bigg] \\ &-\alpha\Bigg[\log\int_{\mathcal{X}\times\mathcal{Y}} e^{(\hat{u}(x)+\hat{v}(y)-\mathsf{c}(x,y))/\alpha}\mathrm{d}PQ(x,y) \\ &\qquad -\log\int_{\mathcal{X}\times\mathcal{Y}} e^{(u^{(l)}(x)+v^{(l)}(y)-\mathsf{c}(x,y))/\alpha}\mathrm{d}PQ(x,y)\Bigg]. \end{aligned} \tag{167}$$

Applying (165) to the second and third terms of (167), and using the fact that P is a probability measure for the first term, yields

$$\mathsf{D}(\hat{u},\hat{v}) - \mathsf{D}(u^{(l)},v^{(l)}) \;\leq\; 2\left\|\hat{u}-u^{(l)}\right\|_{\infty} \;+\; 2\left\|\hat{v}-v^{(l)}\right\|_{\infty}. \tag{168}$$

Applying (143) and (142) to the right-hand side of (168) concludes the proof. ■

Corollary 3.6. *Let $(u^{(l)},v^{(l)})_{l\geq 0}$ be the iterates defined by Algorithm 1, and let $M \triangleq \max\big(2,\, 3\frac{\beta+1}{\alpha+\beta}\big)$. For each $l\geq 0$, define the measure $\pi^{(l)}$ on $\mathcal{X}\times\mathcal{Y}$ through*

$$\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ}(x,y) \;\triangleq\; \exp\left(\frac{u^{(l+1)}(x)+v^{(l)}(y)-\mathsf{c}(x,y)}{\alpha}\right). \tag{169}$$

Then $\pi^{(l)}\in\Pi(P,\star)$ for all $l\geq 0$, and the sequence $\big(\mathrm{d}\pi^{(l)}/\mathrm{d}PQ\big)$ converges to $\mathrm{d}\hat{\pi}/\mathrm{d}PQ$ in $\mathrm{L}^{\infty}(\mathcal{X}\times\mathcal{Y})$ at a linear speed, that is

$$\left\|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ}-\frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right\|_{\infty} \;\leq\; 2\,e^{(2M+2)\|\mathsf{c}\|_{\infty}/\alpha}\,\rho^{\,l}_{(\mathsf{c},\alpha,\beta)}\,d_{\mathrm{H}}\Big(b^{(0)},\hat{b}\Big), \tag{170}$$

where $\rho_{(\mathsf{c},\alpha,\beta)}$ is defined in (104)*. In particular, $\pi^{(l)}$ converges to $\hat{\pi}$ in total variation at the same linear rate.*

Proof: The proof establishes successively that $\pi^{(l)}$ is in $\Pi(P,\star)$, that the relevant exponents are uniformly bounded, and that this bound combined with the linear convergence of the potentials yields (170). Since $u^{(l+1)}=\Phi(v^{(l)})$ in Algorithm 1, the definition (34) of Φ gives, for every $x\in\mathcal{X}$,

$$\begin{aligned}\int_{\mathcal{Y}}\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ}(x,y)\,\mathrm{d}Q(y) &= e^{u^{(l+1)}(x)/\alpha}\int_{\mathcal{Y}} e^{(v^{(l)}(y)-\mathsf{c}(x,y))/\alpha}\,\mathrm{d}Q(y) \\ &= e^{u^{(l+1)}(x)/\alpha}\cdot e^{-u^{(l+1)}(x)/\alpha} \;=\; 1. \end{aligned} \tag{171}$$

Fubini's theorem [4, Theorem 8.8] then implies that $\pi^{(l)}$ is a probability measure with first marginal $(\pi^{(l)})^{\mathcal{X}}_{\#}=P$, hence $\pi^{(l)}\in\Pi(P,\star)$. From (45), it holds that $\|v^{(l)}\|_{\infty}\leq M\|\mathsf{c}\|_{\infty}$ for all $l\geq 1$, and from (44), $\|u^{(l+1)}\|_{\infty}\leq\|v^{(l)}\|_{\infty}+\|\mathsf{c}\|_{\infty}\leq$

$(M+1)\|\mathsf{c}\|_\infty$ for all $l \geq 1$. The same estimates hold for $\hat{u}$ and $\hat{v}$, since $(\hat{u}, \hat{v})$ satisfies (33) and (35). Hence, for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$ and all $l \geq 0$, the two arguments

$$\frac{u^{(l+1)}(x) + v^{(l)}(y) - \mathsf{c}(x,y)}{\alpha} \quad \text{and} \quad \frac{\hat{u}(x) + \hat{v}(y) - \mathsf{c}(x,y)}{\alpha} \tag{172}$$

are both bounded in absolute value by $(2M+2)\|\mathsf{c}\|_\infty/\alpha$. For real numbers a and b with $|a|, |b| \leq R$, the mean value theorem gives $|e^a - e^b| \leq e^R|a-b|$. Applying this pointwise with $R = (2M+2)\|\mathsf{c}\|_\infty/\alpha$ and using (172), for all $(x, y) \in \mathcal{X} \times \mathcal{Y}$ it holds that

$$\begin{aligned} &\left|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ}(x,y) - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}(x,y)\right| \\ &\leq \frac{e^{(2M+2)\|\mathsf{c}\|_\infty/\alpha}}{\alpha}\left(\left|u^{(l+1)}(x) - \hat{u}(x)\right| + \left|v^{(l)}(y) - \hat{v}(y)\right|\right). \end{aligned} \tag{173}$$

Applying (143) and (142) to the previous expression yields

$$\left\|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ} - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right\|_\infty \leq 2\, e^{(2M+2)\|\mathsf{c}\|_\infty/\alpha}\, \rho^l_{(\mathsf{c},\alpha,\beta)}\, d_{\mathrm{H}}\left(b^{(0)}, \hat{b}\right), \tag{174}$$

which gives (170). The total variation bound then follows from

$$\left\|\pi^{(l)} - \hat{\pi}\right\|_{TV} \triangleq \frac{1}{2}\int_{\mathcal{X}\times\mathcal{Y}}\left|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ} - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right| \mathrm{d}PQ \leq \frac{1}{2}\left\|\frac{\mathrm{d}\pi^{(l)}}{\mathrm{d}PQ} - \frac{\mathrm{d}\hat{\pi}}{\mathrm{d}PQ}\right\|_\infty, \tag{175}$$

since PQ is a probability measure. ■

3.6 Experimental results

a venir

References

[1] C. Zalinescu, *Convex analysis in general vector spaces*, 1st ed. Singapore: World Scientific, 2002.

[2] M. Cuturi, "Sinkhorn distances: Lightspeed computation of optimal transportation distances," in *Proceedings of the International Conference on Neural Information Processing Systems (NeurIPS)*, vol. 2, Dec. 2013, pp. 2292–2300.

[3] L. Chizat, G. Peyré, B. Schmitzer, and F.-X. Vialard, "Scaling algorithms for unbalanced transport problems," *Mathematics of Computation*, vol. 87, no. 314, pp. 2563–2609, Feb. 2018.

[4] W. Rudin, *Real and Complex Analysis*, 3rd ed. New York, NY, USA: McGraw-Hill, 1987.

[5] S. Di Marino and A. Gerolin, "Optimal transport losses and sinkhorn algorithm with general convex regularization," *arXiv preprint arXiv:2007.00976*, Jul. 2020.

[6] H. Brézis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, 1st ed. New York, NY: Springer, 2011.

[7] B. Lemmens and R. D. Nussbaum, "Birkhoff's version of hilbert's metric and its applications in analysis," *arXiv: Metric Geometry*, 2013. [Online]. Available: https://api.semanticscholar.org/CorpusID:119321129

[8] R. D. Nussbaum, "Fixed point theorems and denjoy–wolff theorems for hilbert's projective metric in infinite dimensions," *Topological Methods in Nonlinear Analysis*, vol. 29, no. 2, pp. 199–249, 2007.

RESEARCH CENTRE
SOPHIA ANTIPOLIS – MÉDITERRANÉE

2004 route des Lucioles - BP 93
06902 Sophia Antipolis Cedex

Publisher
Inria
Domaine de Voluceau - Rocquencourt
BP 105 - 78153 Le Chesnay Cedex
inria.fr

ISSN 0249-6399